Formulário Teste 2 - Sensores e Sinais

NMec.: ________________ Nome: ____

Transf. de Z

$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n]z^{-n}, z = re^{j\omega}$

Propriedades

$\mathcal{Z}\{ax_1[n] + bx_2[n]\} = aX_1(z) + bX_2(z)$	ROC $\supseteq R_1 \cap R_2$
$\mathcal{Z}\{x[n-n_0]\} = z^{-n_0}X(z)$	ROC R (exceto $z = 0$ ou $z = \infty$)
$\mathcal{Z}\{a^n x[n]\} = X(a^{-1}z)$	ROC $= \|a\|R$
$\mathcal{Z}\{x[-n]\} = X(z^{-1})$	ROC $= R^{-1}$
$\mathcal{Z}\{x^*[n]\} = X^*(z^*)$	ROC $= R$
$\mathcal{Z}\{x_1[n] * x_2[n]\} = X_1(z) \cdot X_2(z)$	ROC $\supseteq R_1 \cap R_2$
$\mathcal{Z}\{x[n] - x[n-1]\} = (1 - z^{-1})X(z)$	ROC $\supseteq R \cap \{\|z\| > 0\}$
$\mathcal{Z}\left\{\sum_{k=-\infty}^{n} x[k]\right\} = \frac{1}{1-z^{-1}}X(z)$	ROC $\supseteq R \cap \{\|z\| > 1\}$
$\mathcal{Z}\{nx[n]\} = -z\frac{dX(z)}{dz}$	ROC $= R$
$x[0] = \lim_{z\to\infty} X(z)$	(Teorema do Valor Inicial)

Pares Fundamentais

$x[n]$	$X(z)$	**ROC**
$\delta[n]$	1	Todo o plano z
$u[n]$	$\dfrac{z}{z-1} = \dfrac{1}{1-z^{-1}}$	$\|z\| > 1$
$a^n u[n]$	$\dfrac{z}{z-a} = \dfrac{1}{1-az^{-1}}$	$\|z\| > \|a\|$
$-a^n u[-n-1]$	$\dfrac{z}{z-a} = \dfrac{1}{1-az^{-1}}$	$\|z\| < \|a\|$
$na^n u[n]$	$\dfrac{az}{(z-a)^2}$	$\|z\| > \|a\|$
$\cos(\omega_0 n)u[n]$	$\dfrac{z(z-\cos\omega_0)}{z^2 - 2z\cos\omega_0 + 1}$	$\|z\| > 1$
$\sin(\omega_0 n)u[n]$	$\dfrac{z\sin\omega_0}{z^2 - 2z\cos\omega_0 + 1}$	$\|z\| > 1$
$a^n \cos(\omega_0 n)u[n]$	$\dfrac{z(z - a\cos\omega_0)}{z^2 - 2az\cos\omega_0 + a^2}$	$\|z\| > \|a\|$

Filtros digitais

$\text{dB} = 10\log_{10}\left(\frac{P_2}{P_1}\right)$

$\text{dB} = 20\log_{10}\left(\frac{A_2}{A_1}\right)$

- Inversão espectral: Mudar o sinal de cada amostra no núcleo do filtro e adicionar um à amostra central
- Reversão espectral: Trocar o sinal amostra-sim/amostra-não

Filtros Windowed-sinc

$h[n] = \frac{\sin(2\pi f_c n)}{n\pi}$, com f_c a frequência de corte normalizada

- Janela de Hamming: $w[n] = 0.54 - 0.46\cos\left(\frac{2\pi n}{M}\right), \quad 0 \le n \le M$
- Janela de Blackman: $w[n] = 0.42 - 0.5\cos\left(\frac{2\pi n}{M}\right) + 0.08\cos\left(\frac{4\pi n}{M}\right), \quad 0 \le n \le M$
- Janela de Hanning: $w[n] = 0.5 - 0.5\cos\left(\frac{2\pi n}{M}\right), \quad 0 \le n \le M$

$M \approx \frac{4}{BW}$

Desenho de filtros por posicionamento de polos-zeros

$BW \approx \frac{(1-r)\cdot f_s}{\pi}$

$|H_{\text{peak}}| \approx \frac{1}{2(1-r)}$

$(z - re^{j\theta})(z - re^{-j\theta}) = z^2 - 2r\cos(\theta)z + r^2$

$Q = \frac{f_c}{BW}$

$|H(e^{j\omega})| = K\frac{\prod_{j=1}^{M}|e^{j\omega}-z_j|}{\prod_{i=1}^{N}|e^{j\omega}-p_i|}$

$\angle H(e^{j\omega}) = \sum_{j=1}^{M}\angle(e^{j\omega} - z_j) - \sum_{i=1}^{N}\angle(e^{j\omega} - p_i)$

PB - $H(z) = K\frac{(z+1)^2}{(z-p_1)(z-p_2)}$

PA - $H(z) = K\frac{(z-1)^2}{(z-p_1)(z-p_2)}$

PBand - $H(z) = K\frac{1-z^{-2}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$, com

$Q \approx \frac{\pi f_0}{(1-r)f_s}$

Notch - $H(z) = \frac{1-2\cos(\omega_0)z^{-1}+z^{-2}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$

ADC e DACs

$LSB = \frac{V_{ref}^{+}-V_{ref}^{-}}{2^N-1}$

Erro Máximo $= \pm\frac{1}{2}LSB$

$SNR_{AD} \approx 6,02N + 1,76$ dB

$SNR_{AD} = 6,02N + 1,76 \text{ dB} + 10\log_{10}\left(\frac{f_s}{2\times BW}\right)$

$DR = 2^N - 1$

$DR_{\text{dB}} \approx 6,02N + 1,76$ dB

$ENOB = \frac{SNR_{\text{real}}(\text{dB})-1,76}{6,02}$

Atraso de Grupo $= \frac{36}{f_s}$

Tempo Total de Estabilização do Filtro $= \frac{72}{f_s}$

$V_{out} = V_{ref} \times$ Código Digital

Sensores

$S = \frac{\Delta V_{out}}{\Delta X_{in}}$

$\Delta X_{min} = \frac{\text{Amplitude}}{2^n-1}$

Lin $= \frac{\text{Desvio máx.}}{\text{Escala total}} \times 100\%$

$X_{corrigido} = a_0 + a_1X_{raw} + a_2X_{raw}^2 + a_3X_{raw}^3 + \ldots$

Gain (dB) $= 20\log_{10}\left(\frac{V_{out}}{V_{in}}\right)$

$y(t) = y_{final}(1 - e^{-t/\tau})$

$DR_{dB} = 20\log_{10}\left(\frac{V_{max}}{V_{min}}\right)$

$DR_{dB} \approx 6.02n + 1.76$

$\sigma_q = \frac{q}{\sqrt{12}} = \frac{LSB}{\sqrt{12}}$

$V_{n,total} = \sqrt{V_{thermal}^2 + V_{1/f}^2 + V_{shot}^2 + V_{quant}^2}$

$SNR_{dB} = 20\log_{10}\left(\frac{V_{signal}}{V_{noise}}\right)$

Ganho $SNR = 10\log_{10}(OSR)$ dB, com

$OSR = \frac{f_s}{2\times f_{signal\_BW}}$

$TC = \frac{\frac{\Delta X}{X_{nominal}}}{\Delta T}$

$V_{compensated} = \frac{V_{measured}-V_{offset}(T)}{G(T)}$

Histerese $= \frac{\text{Diferença máx.}}{\text{Escala total}} \times 100\%$

$y(t) = \int\int \mu(\alpha,\beta)\hat{\gamma}_{\alpha,\beta}[x(t)]\,d\alpha\,d\beta$

Headroom (dB) $= 20\log_{10}\left(\frac{V_{max}}{V_{nominal}}\right)$

$CF = \frac{V_{peak}}{V_{RMS}}$

Carga (%) $= T_{process} \times f_s \times 100$

SI, Medidas e Erros

$\Delta y = \sum_i \left|\frac{\partial f}{\partial x_i}\right| |\Delta x_i|$

$s_y^2 = \left(\frac{\partial f}{\partial x_1}\right)^2 s_{x_1}^2 + \left(\frac{\partial f}{\partial x_2}\right)^2 s_{x_2}^2 + \ldots$

Probabilidades para uma v.a. $X \sim \mathcal{N}(0,1)$

X	$P[X < x]$	$P[\|X\| < x]$
0.000	0.500	0.000
1.000	0.841	0.683
2.000	0.977	0.954
3.000	0.999	0.997
4.000	1.000	1.000